Quadratic Likelihood

Matias Shedden

Distribution of the MLE

Consider a general setting where we have data (some collection of random variables). In this class, that has generally meant a random sample $X_1, \cdots, X_n$, all independent and identically distributed. This need not be the case in general, however. We only need to have that the distribution of the entire random vector $(X_1, \cdots, X_n)$ is parameterized by some parameter θ.

Consider the situation in which the distribution of $(X_1, \cdots, X_n)$ is parameterized by a one-dimensional parameter value θ (if there are other parameters in the model, we can think of them as fixed and known). Then the random sample has some joint pdf, i.e. the likelihood:

$$f_\theta(x) = f_\theta(x_1, \cdots, x_n) = L(\theta)$$

where we let $f_\theta(x)$ denote the joint pdf of our data corresponding to the parameter value θ.

In a similar vein, we will introduce the following notation. Suppose that $(X_1, \cdots, X_n) \sim \text{fam}(\theta)$ (the random sample is drawn from some distribution with parameter θ). Then

$$E_\theta(g(X_1, \cdots, X_n)) = \int \cdots \int g(x_1, \cdots, x_n) f_\theta(x_1, \cdots, x_n) dx_1 \cdots dx_n$$

That is, E_θ denotes expected value taken when the parameter value is equal to θ. If the parameter value was equal to some "true value" θ_0, for example, we would denote that expectation as E_{θ_0}, and the corresponding density function as f_{θ_0}

Suppose that the log-likelihood is exactly quadratic in θ, that is

$$\ell(\theta) = \log(L(\theta)) = U + \theta Z - \frac{1}{2} K \theta^2$$

where U and Z are arbitrary functions of our data $(X_1, \cdots, X_n)$, and K is a constant.

If we wish to find the MLE for θ, we can take the derivative and set it equal to 0:

$$\frac{\partial \ell}{\partial \theta} = Z - K\theta \stackrel{set}{=} 0 \quad \Longrightarrow \quad \hat{\theta} = Z/K$$

Now consider two possible values of our parameter: θ, and $\theta + \delta$. Then we may write the difference in the log-likelihood evaluated at these two parameter values as:

$$\begin{aligned} \ell(\theta + \delta) - \ell(\theta) &= U + (\theta + \delta)Z - \frac{1}{2}K(\theta + \delta)^2 - U - \theta Z + \frac{1}{2}K\theta^2 \\ &= \delta Z - \frac{1}{2}K\theta^2 - K\theta\delta - \frac{1}{2}K\delta^2 + \frac{1}{2}K\theta^2 \\ &= (Z - K\theta)\delta - \frac{1}{2}K\delta^2 \end{aligned}$$

We now note the following:

$$\begin{aligned} E_\theta(e^{\ell(\theta+\delta)-\ell(\theta)}) &= \int e^{\ell(\theta+\delta)-\ell(\theta)} f_\theta(x)dx \\ &= \int \frac{f_{\theta+\delta}(x)}{f_\theta(x)} f_\theta(x)dx \\ &= \int f_{\theta+\delta}(x)dx \\ &= 1 \end{aligned}$$

which implies

$$\begin{aligned} E_\theta(e^{\ell(\theta+\delta)-\ell(\theta)}) &= 1 \\ E_\theta(e^{(Z-K\theta)\delta-\frac{1}{2}K\delta^2}) &= 1 \\ E_\theta(e^{Z\delta}) &= e^{K\theta\delta+\frac{1}{2}K\delta^2} \end{aligned}$$

Recall that δ is a constant, so this is exactly showing that the moment generating function of Z, $M_Z(\delta) = E_\theta(e^{Z\delta})$ is equal to the moment generating of a normal random variable. Recall that for $X \sim N(\mu, \sigma^2)$,

$$M_X(\delta) = e^{\mu\delta+\frac{1}{2}\sigma^2\delta^2}$$

Hence we have proven that

$$Z \sim N(K\theta, K)$$

which implies that the MLE has distribution

$$\hat{\theta} = Z/K \sim N\left(\theta, \frac{1}{K}\right)$$

This highlights the general principle of regarding the approximate distribution of the MLE:

Approximate Distribution of the MLE: If the log-likelihood $\ell(\theta)$ is approximately quadratic in θ, with K constant, then the distribution of the MLE for θ is approximately normal.

Asymptotic Justification

We now consider the question of how one can, in general, justify the statement "$\ell(\theta)$ is approximately quadratic in θ". For the above to be true, we need the following:

- $\ell(\theta)$ is approximately quadratic
- K is approximately constant (not random)

We will return to the setting that we consider most commonly in this class, where $X_1, \cdots, X_n$ is a *random sample* (independent and identically distributed) from some distribution family parameterized by θ. We will show that the above approximation is justified as $n \to \infty$. That is, we will use distributional results to show that $\ell(\theta)$ is equal to a quadratic in θ with constant K, plus a term which converges to 0 as $n \to \infty$. This *asymptotic* result is what we will call our *approximation* in finite samples.

We will suppose that we have observed a random sample from a distribution with true parameter value θ_0. We do this because our asymptotic justification for the quadratic approximation will only hold *near* the true parameter value θ_0.

We will redefine θ by rescaling by $\sqrt{K}$ and subtracting θ_0. That is, we will show that, for a *particular choice of* K,

- $\ell(\theta)$ is approximately quadratic in $\sqrt{K}(\theta - \theta_0)$
- The quantity in front of the quadratic term is approximately constant

We start by making the following important assumption.

Assumption: ($\sqrt{n}$-consistency): Let $\hat{\theta}$ be the MLE of θ. Assume that

$$\sqrt{nI(\theta_0)}(\hat{\theta} - \theta_0) = O_p(1)$$

where $I(\theta)$ is the *Fisher Information* (for one observation):

$$I(\theta) = V_\theta \left(\frac{\partial}{\partial \theta} \log(f_\theta^1(X)) \right)$$

$f_\theta^1(x)$ is used here to represent the density of X_1 (a single element of the random sample). $I(\theta)$ is a constant (because the variance of a random variable is just a constant).

The constant $\sqrt{nI(\theta_0)}$ will prove to be very important in the upcoming proof. Note the above means that for every $\epsilon > 0$, there exists a constant $M < \infty$ with

$$P\left(\left| \sqrt{nI(\theta_0)}(\hat{\theta} - \theta_0) \right| \leq M \right) \geq 1 - \epsilon.$$

That is,

$$P\left(\left| \hat{\theta} - \theta_0 \right| \leq \frac{M}{\sqrt{nI(\theta_0)}} \right) \geq 1 - \epsilon,$$

which, in words, means that for arbitrary choice of ϵ, *there exists* M *such that the probability that* $\hat{\theta}$ *is within* $M/\sqrt{nI(\theta_0)}$ *of* θ_0 *occurs with probability at least* $1 - \epsilon$. This is important, because it is precisely in this *neighborhood* of θ_0 (an open set containing θ_0) that our quadratic approximation of the likelihood will hold.

We now consider our log-likelihood $\ell(\theta)$, a function of our parameter value θ. We perform a second-order Taylor expansion of the log likelihood:

$$\ell(\theta) = \ell(\theta_0) + \ell'(\theta_0)(\theta - \theta_0) + \frac{1}{2}\ell''(\theta_0)(\theta - \theta_0)^2 + R$$

where

$$R = R_n = \frac{1}{6}(\theta - \theta_0)^3 \ell'''(\tilde{\theta})$$

for some value $\tilde{\theta}$ between θ and θ_0.

Fix $\epsilon > 0$. Due to the $\sqrt{n}$-consistency assumption, we know that there exists M such that the maximum of $\ell(\theta)$ (i.e. $\hat{\theta}$) belongs to the following set with probability at least $1 - \epsilon$:

$$U_{n,M} = \left[\theta_0 - \frac{M}{\sqrt{nI(\theta_0)}}, \theta_0 + \frac{M}{\sqrt{nI(\theta_0)}} \right]$$

Let $\theta \in U_{n,M}$. Clearly, $\theta_0 \in U_{n,M}$ (its at the center of the interval), so any $\tilde{\theta}$ which lies between θ and θ_0 will also belong to $U_{n,M}$. Therefore, by the definition of inclusion in the set $U_{n,M}$,

$$\left|\sqrt{nI(\theta_0)}(\tilde{\theta}-\theta_0)\right| \leq M$$

and hence

$$|R| = \frac{1}{6}(\theta-\theta_0)^3|\ell'''(\tilde{\theta})| \leq \frac{1}{6}\left(\frac{M}{\sqrt{nI(\theta_0)}}\right)^3 |\ell'''(\tilde{\theta})|$$

Note that $\ell'''(\theta)$ is just the sum of n random variables:

$$\ell'''(\theta) = \sum_{i=1}^{n} \frac{\partial^3}{\partial\theta^3}\log(f_\theta(X_i))$$

so it grows at a rate of n (in a random sense, provided the random variable $\frac{\partial^3}{\partial\theta^3}\log(f_\theta(X_i))$ has finite expected value. We omit the details here). A sequence of random variables growing at a rate of n divided by $n^{3/2}$ (which grows faster than n) will converge in probability to 0. Hence,

$$|R| = \frac{M^3}{6I(\theta_0)^{3/2}}\frac{|\ell'''(\tilde{\theta})|}{n^{3/2}} \xrightarrow{P} 0$$

There are two probabilistic statements being made here: there exists M such that $\hat{\theta}$ belongs to the set $U_{n,M}$ with probability at least $1-\epsilon$, and, for fixed M, on that set, R converges in probability to 0. The single statement combining the two would be to say that for any $\epsilon > 0$, there exists $M < \infty$ such that for every $\rho > 0$,

$$P(\hat{\theta} \in U_{n,M} \text{ and } |R| \leq \rho) \geq 1-\epsilon-o(1)$$

Here, $o(1)$ denotes a sequence converging to 0. That is, $\hat{\theta}$ lies in the neighborhood, and the remainder term is smaller than ρ, with probability greater than or equal to $1-\epsilon$ minus a term converging to 0.

We will allow this to be enough to justify to us that as $n \to \infty$, we can consider $\ell(\theta)$ to be *approximately* quadratic in the neighborhood of θ_0 we defined. Remember that this neighborhood wasn't defined arbitrarily: it was the neighborhood in which the MLE is guaranteed to belong with arbitrarily high probability. Ignoring R, we can rewrite the log-likelihood as

$$\ell(\theta) = \ell(\theta_0) + \frac{1}{\sqrt{K}}\ell'(\theta_0)\sqrt{K}(\theta-\theta_0) + \frac{1}{2}\frac{1}{K}\ell''(\theta_0)(\sqrt{K}(\theta-\theta_0))^2$$

Letting $\theta^* = \sqrt{K}(\theta-\theta_0)$, $U = \ell(\theta_0)$, and $Z = \frac{1}{\sqrt{K}}\ell'(\theta_0)$, we get

$$\ell(\theta) = U + Z\theta^* + \frac{1}{2}\frac{1}{K}\ell''(\theta_0)(\theta^*)^2$$

which is a quadratic function of θ^*, with U and Z being functions of our random sample $X_1, \cdots, X_n$. The only thing that remains to be shown is that the following can be considered a constant:

$$\frac{\ell''(\theta_0)}{K}$$

For this to be the case, we must choose K in a particular way.

Result:

$$E_\theta\left(-\ell_1''(\theta)\right) = I(\theta)$$

where $\ell_1(\theta) = \log(f_\theta^1(x))$. (Proof omitted here)

Therefore, by the law of large numbers,

$$\frac{1}{n}\ell''(\theta) = \frac{1}{n}\frac{\partial^2}{\partial\theta^2}\sum_{i=1}^{n}\ell_1(\theta|X_i) = \frac{1}{n}\sum_{i=1}^{n}\ell_1''(\theta|X_i) \xrightarrow{P} -I(\theta)$$

We now return to our above expression, and allow $K = nI(\theta_0)$. Then

$$\frac{\ell''(\theta_0)}{K} = \frac{\frac{1}{n}\ell''(\theta_0)}{\frac{1}{n}K} = \frac{\frac{1}{n}\ell''(\theta_0)}{I(\theta_0)} \xrightarrow{P} -1$$

Therefore we may rewrite our likelihood as

$$\ell(\theta) = U + Z\theta^* + \frac{1}{2}\frac{\ell''(\theta_0)}{K}(\theta^*)^2 = U + Z\theta^* - \frac{1}{2}(\theta^*)^2 + \frac{1}{2}(\theta^*)^2\left(\frac{\ell''(\theta_0)}{K} - (-1)\right)$$

where the last term in the above converges in probability to 0. Therefore, as $n \to \infty$, $\ell(\theta)$ is equivalent to the following with high probability:

$$\ell(\theta) = U + Z\theta^* - \frac{1}{2}K^*(\theta^*)^2$$

where $K^* = 1$. This is a quadratic function of θ^*, where θ^* is bounded ($|\theta^*| \leq M$) on the neighborhood $U_{n,M}$ that contains $\hat{\theta}$ with probability at least $1 - \epsilon$. Thus, our earlier results imply

$$Z \overset{approx}{\sim} N(K^*\theta^*, K^*) = N(\sqrt{nI(\theta_0)}(\theta - \theta_0), 1)$$

and

$$\sqrt{nI(\theta_0)}(\hat{\theta} - \theta_0) = Z/K^* \overset{approx}{\sim} N(\sqrt{nI(\theta_0)}(\theta - \theta_0), 1)$$

where "approx" indicates that our likelihood was not exactly quadratic, but rather approximately so.

Note that the parameter value θ that we are estimating is equal to θ_0, so the above is simply

$$\sqrt{nI(\theta_0)}(\hat{\theta} - \theta_0) \overset{approx}{\sim} N(0, 1)$$

which is the classic distributional result concerning the MLE.

Note that we refrained from plugging in $\theta = \theta_0$ at earlier steps of the proof because in order to maximize the likelihood, θ must be considered as an unknown variable, which we take a derivative with respect to. When applying maximum likelihood estimation, we never know θ_0. Therefore, we usually make statements in the following form:

Suppose that $\theta = \theta_0$. In the event that this is true (under the *hypothesis* that $\theta = \theta_0$),

$$\sqrt{nI(\theta_0)}(\hat{\theta} - \theta_0) \overset{approx}{\sim} N(0, 1)$$

or, in an asymptotic phrasing,

$$\sqrt{nI(\theta_0)}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, 1)$$

For example, we might say suppose that the true value of $\theta = 5$. Then under that *hypothesis*,

$$\sqrt{nI(5)}(\hat{\theta} - 5) \overset{approx}{\sim} N(0, 1)$$

If that hypothesis is not true, we have no guarantee that the approximation will hold. In fact, we can construct approximations that show that if the hypothesis is false, the distribution will be approximately normal with a different mean, or not normal at all.

Furthermore, our consistency assumption implies that $\hat{\theta} \xrightarrow{P} \theta_0$. If we additionally assume that $I(\theta)$ is a continuous function of θ, the continuous mapping theorem and Slutsky's theorem imply that

$$\sqrt{nI(\hat{\theta})}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, 1)$$

References

Geyer, C. J. (2013). Asymptotics of maximum likelihood without the LLN or CLT or sample size going to infinity. Advances in Modern Statistical Theory and Applications: A Festschrift in Honor of Morris L. Eaton (Vol. 10, pp. 1–24). Institute of Mathematical Statistics. https://doi.org/10.1214/12-IMSCOLL1001